

GTU CSE 241 / 501 Quiz

Std Code: **106**

Std Number:

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```
#include <iostream>
using namespace std;
#include <memory>
#include <stdexcept>

class Set {
private:
    int *data;
    int size;
    int capacity;
public:
    shared_ptr<Set> ptr(900);
    make_shared<Set> ptr(data);

    Set(int nsize, int ncap): size(nsize), capacity(ncap),
    data(nullptr) {}

    ~Set() {
        delete []data;
    }

    Set(Set& o): size(o.size), capacity(o.capacity) {
        delete []data;
        data = new int[size];
        data = o.data;
    }

    Set& operator =(Set& o) {
        if (&this != &o) {
            delete []data;
            size = o.size;
            capacity = o.capacity;
            data = new int[size];
            for (int i = 0; i < size; i++) {
                data[i] = o.data[i];
            }
        }
        return *this;
    }
}
```

```

    set(set && o) noexcept : size(o.size), capacity(o.capacity) {
        data = o.data;
        o.data = nullptr;
        o.size = 0;
        o.capacity = 0;
    }

```

```

set& operator=(set&& o) noexcept {
    if(&this != &o) {
        delete [] data;
        size = o.size;
        capacity = o.capacity;
        data = o.data;
        o.data = nullptr;
        o.size = 0;
        o.capacity = 0;
    }
    return *this;
}

```

```

void AddElement(int value) {
    if(capacity == 0) {
        throw out_of_range("Don't have memory");
    }
    if(size <= capacity) {
        capacity *= 2;
        data = new int[capacity];
    }
    data[size++] = value;
}

```

```

int getS() const { return size; }
for(int i = 0;
bool doesContain(int value) {
    for(int i = 0; i < getS(); i++) {
        if(data[i] == value) {
            return true;
        }
    }
    return false;
}
} // close set class

```